

CubeCoach Project Report

Group 12

January 5, 2026

1 A: Description of Your Key Contribution

As a member of the CubeCoach project Web App sub-team, my primary contribution was the design and implementation of the user interface (UI) for the web application, including the development of the "Onboarding Tutorial" feature, which helps new users learn how to use the application and understand the Rubik's Cube move notation. This work was critical to the overall project, as the Web App serves as the primary interface through which users interact with the system, directly impacting their experience and learning effectiveness.

1.1 Implementation Approach

Based on research into the challenges faced by beginners learning the Rubik's Cube, I found that most users struggle to correlate the cube's physical state with the tutorial's visual representation. To address this issue, I designed the following features:

- **Clear layout structure:** The interface was divided into cube visualization area, step guidance area, and control panel, making it easy for users to find the features they need. The figure below shows the final implemented interface layout:
- **Onboarding Tutorial:** Based on feedback from Demo 2, I developed an interactive tutorial that guided users through the interface features and Rubik's Cube move notation. Key components of the tutorial include:
 - Introduction to application features, including 3D visualization controls and step navigation
 - Explanation of the cube notation system (F, R, L, B, U, D and their variations)
 - Interactive exercises allowing users to master the basics before proceeding
- **Responsive Design:** The web app was optimized for use on various devices (desktop and mobile), ensuring a seamless user experience across platforms.

Technologically, I used the Next.js framework for development and Vercel for automated deployment to ensure zero-downtime updates. I collaborated closely with team member Andreas to integrate the 3D Rubik's Cube visualization, ensuring a smooth interface and functional integration.

1.2 Success Measurement

We assessed the success of the UI through multiple methods:

- **Lighthouse Test:** Our Web App received excellent scores for performance, accessibility, best practices, and SEO (performance: 95, accessibility: 100, best practices: 100, SEO: 100). This demonstrates that the interface I developed is not only aesthetically pleasing but also complies with modern web development standards and accessibility requirements.
- **User Feedback:** In Demo 2, a user study involving 10 participants showed that 9 out of 10 agreed the tool was easy to use. All participants believed the system's functions were well-integrated and felt confident using the tool. Below is some of the user feedback data:

Evaluation Statement	Agree	Neutral	Disagree
I found this tool easy to use	9	1	0
I found the various functions in this tool were well integrated	10	0	0
I felt confident using this tool	10	0	0

Table 1: User Experience Survey Results (10 participants)

- **Iterative Improvements:** Based on user feedback, I made several improvements to the UI, most importantly implementing a "relative mode" for the cube instructions, where the directions are always relative to the user's current view of the cube's front face. This enhancement significantly improved the user experience by addressing difficulties in understanding instructions.

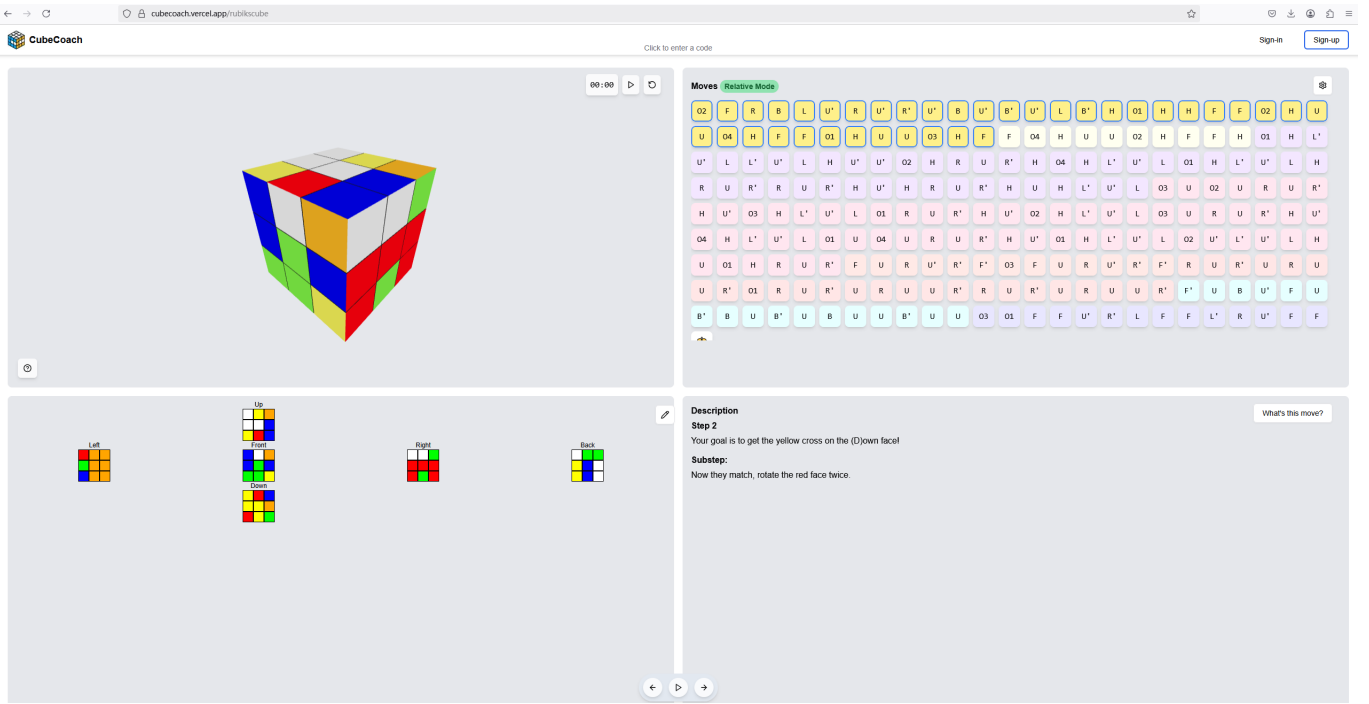


Figure 1: <https://cubecoach.vercel.app/rubikscube>

1.3 Collaboration with the Project

My work closely integrated with other sub-teams:

- **Collaboration with the Algorithms Sub-Team:** I worked closely with the algorithms team to translate the solving algorithms into visual guides for each step. This included processing algorithm data structures and presenting them in a user-friendly format.
- **Coordination with the Computer Vision Sub-Team:** I designed the interface to accurately display the cube's state from the scanning system and provide real-time feedback as users modify the cube. I worked with the vision team to define data exchange formats and communication protocols, the result of the scanning is sent to web app as .js file.
- **Integration with the Manipulation Sub-Team:** Our Web App needed to communicate with the physical robot, sending instructions and receiving status updates. I implemented the necessary API client code to ensure the Web App could communicate correctly with the robot system.

These integrations required frequent communication with team members to ensure all components worked seamlessly together. I also participated in the system's integration testing during the later stages of the project to ensure the Web App functioned correctly with other subsystems.

Through this work, I contributed a fully functional, user-friendly web interface to the CubeCoach project, enabling users to easily learn Rubik's Cube solving techniques and interact with the physical robot system.

2 B: Analysis of What You Learned

2.1 Technical Skill Acquisition

Through the development of the CubeCoach project, I gained improvements in multiple technical areas:

- **Next.js Framework Application:** I deepened my understanding of the Next.js framework, particularly its automated deployment capabilities. As stated in the Demo 1 report, we "used the Next.js framework to develop the Web App, utilizing Vercel's automated deployment for seamless, zero-downtime updates," which greatly improved our development efficiency.
- **React Interactive Component Development:** I mastered the skills for building interactive web interfaces, particularly in implementing 3D cube visualization and editing functionality. The Demo 3 report mentioned that I implemented "more fluid manual editing capabilities," which required me to understand React's state management and user interaction handling mechanisms.

- **Web Application Performance Optimization:** By evaluating Web App performance through the Lighthouse testing tool, I learned to identify and resolve performance bottlenecks. As shown in the Demo 3 report, our application achieved high scores in performance, accessibility, and best practices (95 for performance, 100 for accessibility).

2.2 User Experience Design Skills

In this project, I also enhanced my user experience design capabilities:

- **Tutorial Design:** I designed and implemented two key tutorials—the "Getting Started Tutorial" and the "Cube Notation Tutorial." These tutorials were developed based on user testing feedback from Demo 3 and significantly lowered the user learning curve.
- **User Testing and Feedback Integration:** I learned how to collect and analyze user feedback and transform it into actual features. For example, the Demo 3 report mentioned that based on user feedback, I developed the "relative mode" feature, making instructions always relative to the user's current view of the cube's front side.
- **Accessibility Design:** I learned the importance of web accessibility and applied it in practice, enabling our Web App to achieve a perfect accessibility score of 100, ensuring that a wider range of users could use our application.

2.3 Challenges Encountered and Solutions

During development, I encountered and resolved several key challenges:

- **UI and Backend Integration:** Initially, integrating the web interface with the algorithm and computer vision subsystems was challenging. I successfully implemented seamless communication between systems by designing clear data structures and API interfaces.
- **User Instruction Comprehension Difficulties:** As mentioned in the Demo 3 report, "some users found the move instructions somewhat confusing, as they weren't always relative to the current front face." To solve this problem, I implemented the "relative mode" feature, which greatly enhanced the user experience.
- **Supporting Different Devices:** Ensuring the application ran well on both desktop and mobile devices was a challenge. I adopted responsive design principles to make the interface adaptable to different screen sizes, providing users with a consistent experience.

2.4 Reflections for Future Work

Based on this project experience, I have developed clearer plans for the future:

- **Earlier User Testing:** If starting the project again, I would conduct user testing in the early development stages to collect feedback that would guide design decisions and avoid major changes later.
- **Earlier Definition of Interface Specifications:** Agreeing on data formats and interaction methods with the algorithm and computer vision teams earlier would help reduce workload and issues during the integration phase.
- **Establishing Unified Front-End Coding Standards:** As I mentioned in my report, "as the project progressed, my code style conflicted with Andreas's, leading to some code refactoring." Establishing unified coding standards and architecture early on would improve collaboration efficiency.

2.5 Conclusion

The CubeCoach project not only enhanced my technical abilities but also helped me understand the importance of user experience design and the complexity of cross-team collaboration. These experiences will positively impact my future software development work.